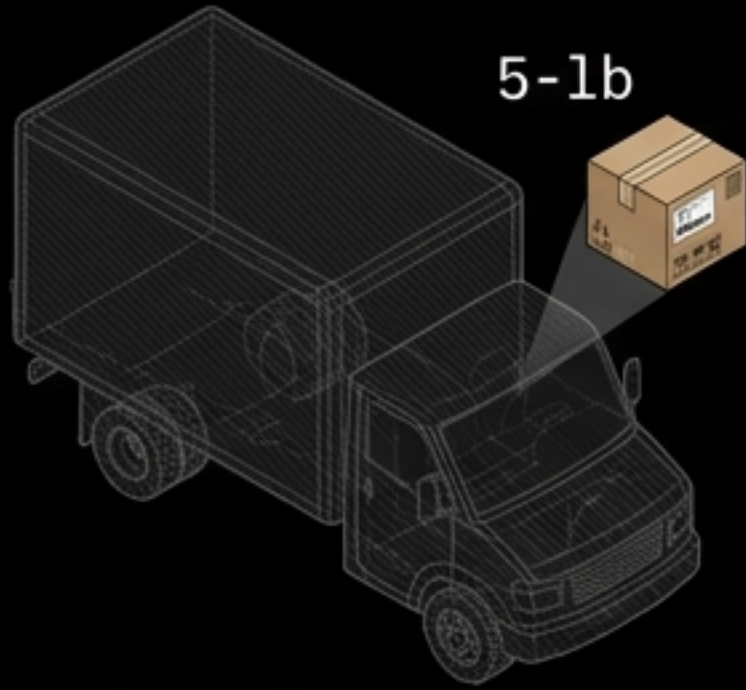


Bypassing the Physics Inefficiency of the Last Mile

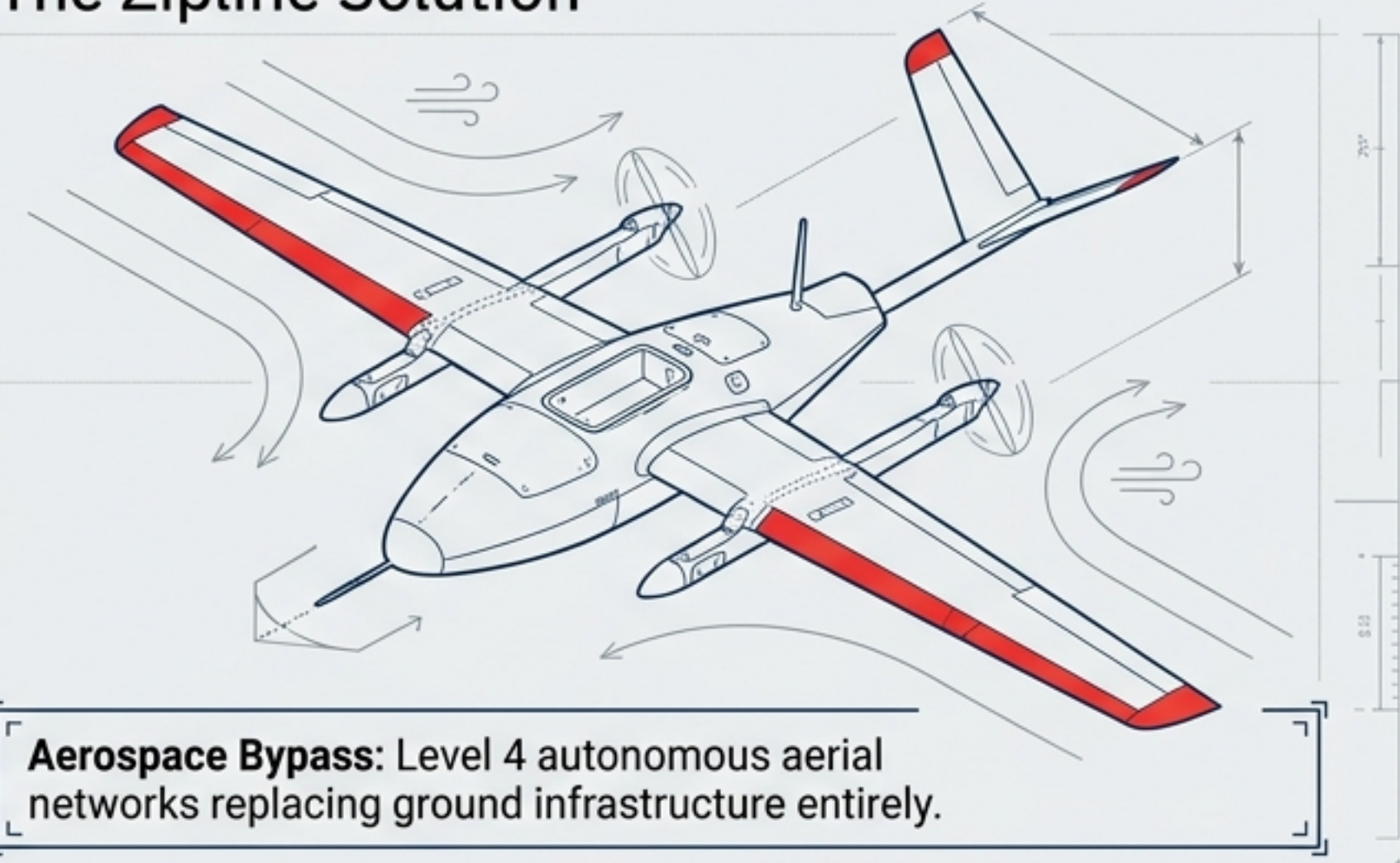
The Broken Paradigm



Weight-to-Payload
Inefficiency Ratio:
3000:5

Traditional Friction: Vulnerable road networks, high carbon emissions, and highly variable transit times.

The Zipline Solution



Aerospace Bypass: Level 4 autonomous aerial networks replacing ground infrastructure entirely.

The Catalyst

5.5 million children die annually due to lack of access to basic medical products.

The Objective

Build the world's first logistics system serving all people equally, bypassing ground infrastructure.

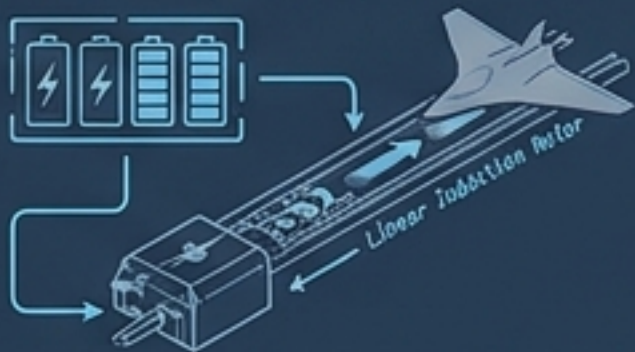
The Execution

120M+ autonomous flight miles flown, proving the viability of massive-scale instant logistics.

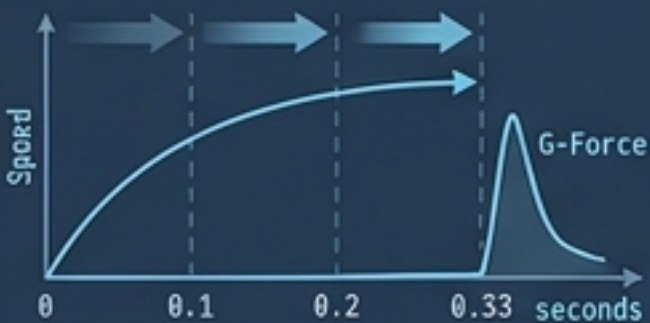
Platform 1 Architecture and the Naval Carrier Launch Model

Launch Physics: Electric Catapult

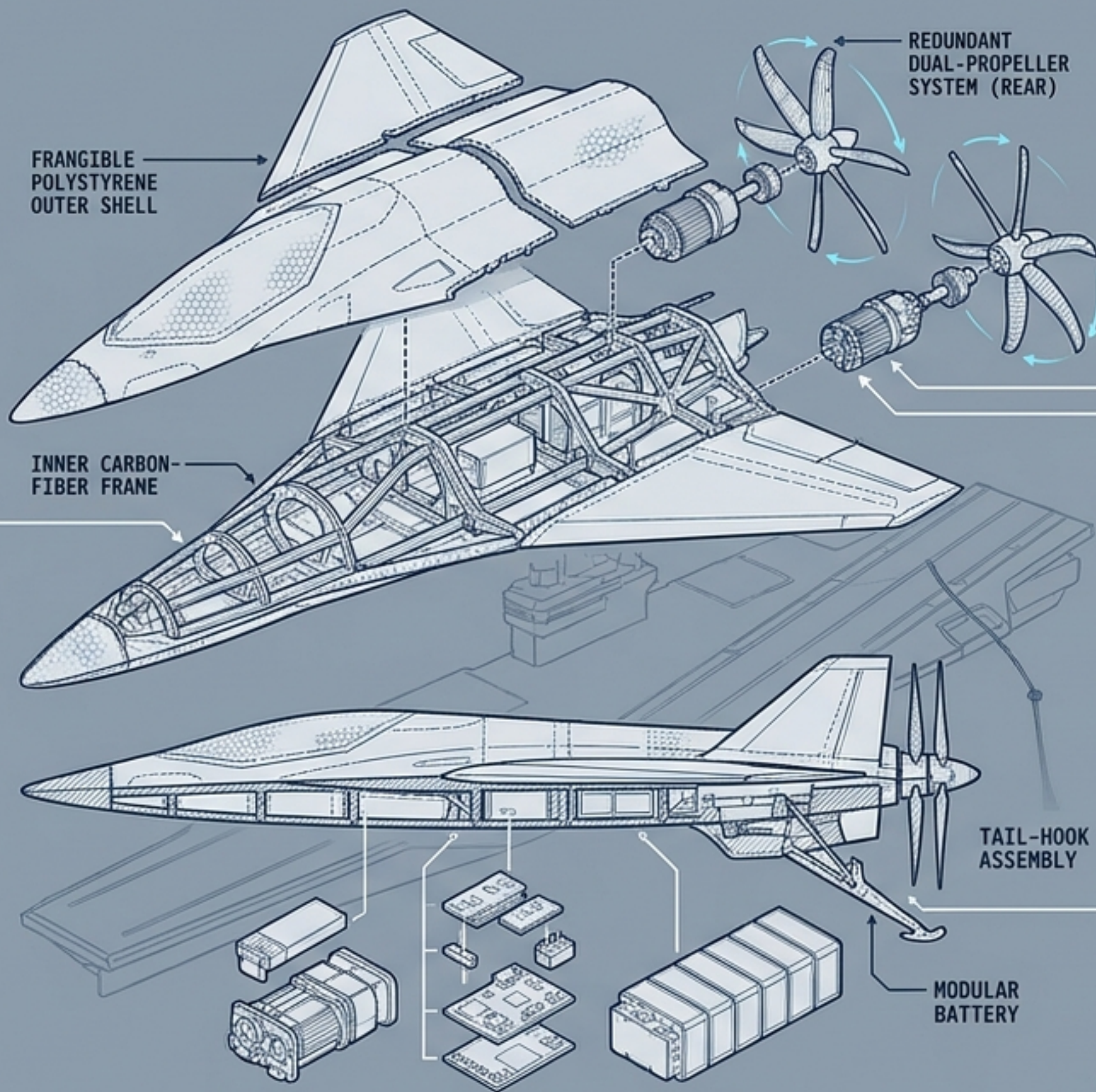
Propulsion:
Supercapacitor-powered electric catapult system.



Acceleration:
0 to 67 mph (108 km/h) in 0.33 seconds.



Efficiency:
Reaches optimal cruising speed instantly, preserving battery for the 100km (62mi) maximum range.



Arrest & Recovery: Wire Snag

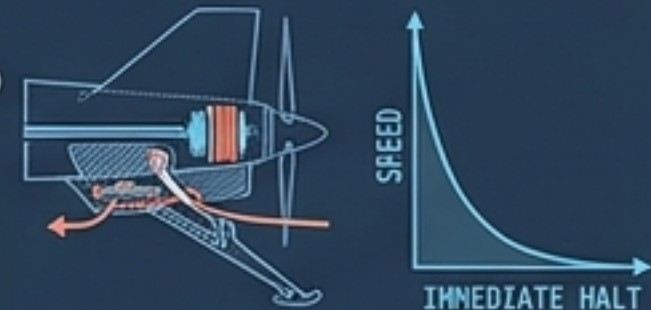
Targeting:
Military-grade GPS guides aircraft return with centimeter-level precision.



Capture:
Autorelated poles swing a capture cable upward in the final second of approach.



Deceleration:
Tail-hook snags the wire, engaging a friction-based braking system for an immediate, runway-free halt.



Platform 2: The eVTOL Mothership and Tethered Droid System

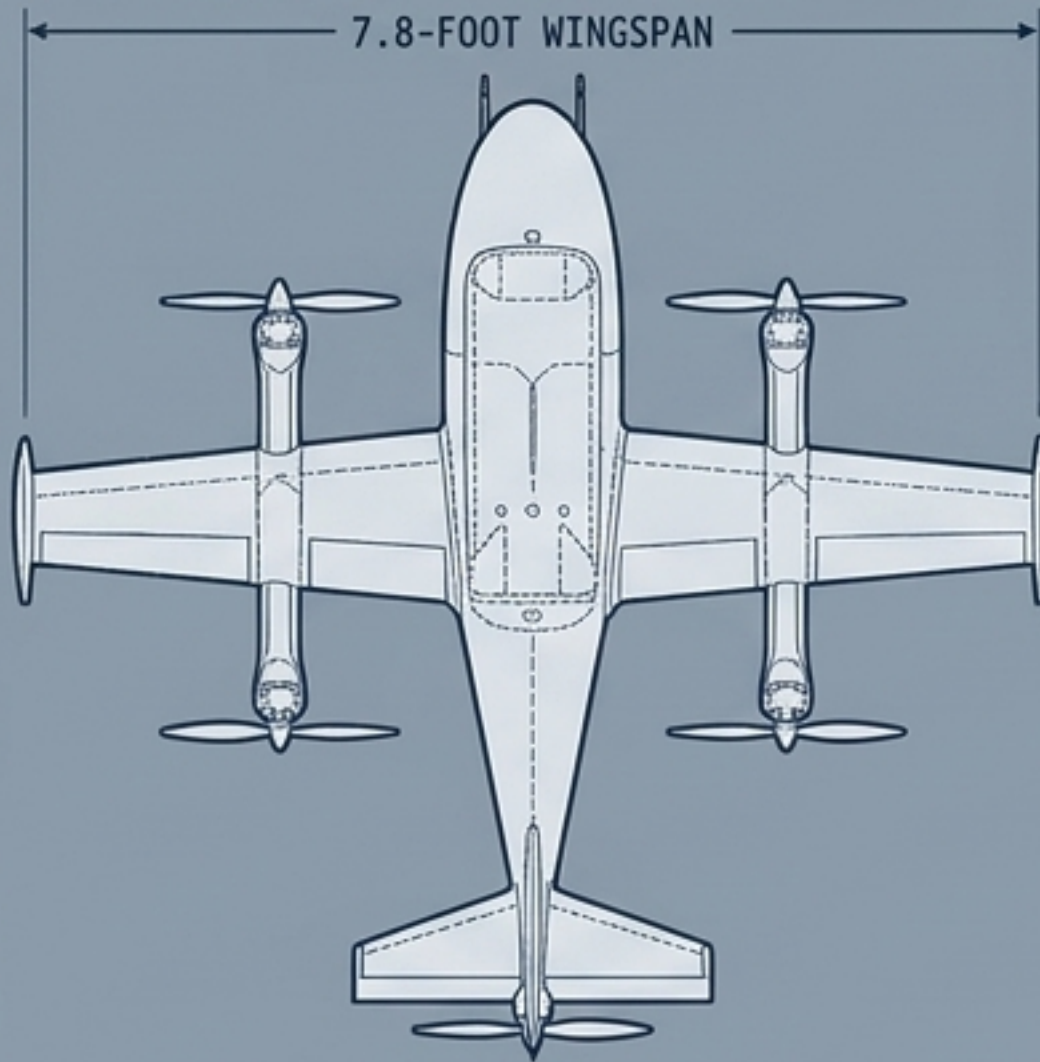
TELEMETRY

55-lb Hybrid eVTOL

70 mph Cruise Speed

8-lb Payload Capacity

10-Mile Hub-and-Spoke Radius



Delivery Sequence

01

Acoustic Stealth Hover

Mothership maintains position at 330 ft AGL. Sound signature at ground level is minimized to the volume of rustling leaves.

02

Active Stabilization Descent

The droid is lowered via tether. Three internal fan thrusters actively counter wind shear during descent, offloading positioning work from the mothership.

03

Precision Release

Bottom doors open for dinner-plate (1-meter) accuracy payload release. Total hover time is strictly capped at 75 seconds.

High-Frequency Distribution Hubs: The 90-Second Launch Cycle



African Proving Ground: Optimizing the Rwandan Blood Supply

The Terrain Challenge: Mountainous geography and poor roads made ground transport unpredictable, necessitating an aerial bypass.



System Health

System Health

Coverage Optimization

>75%

Of the national blood supply outside Kigali is now delivered by Zipline.

System Health

System Health

Velocity Gains

61%

Reduction in blood product delivery times across rural healthcare facilities.

System Health

System Health

Waste Reduction

67%

Decrease in blood unit expirations and total medical spoilage.

System Health

System Health

Emergency Response

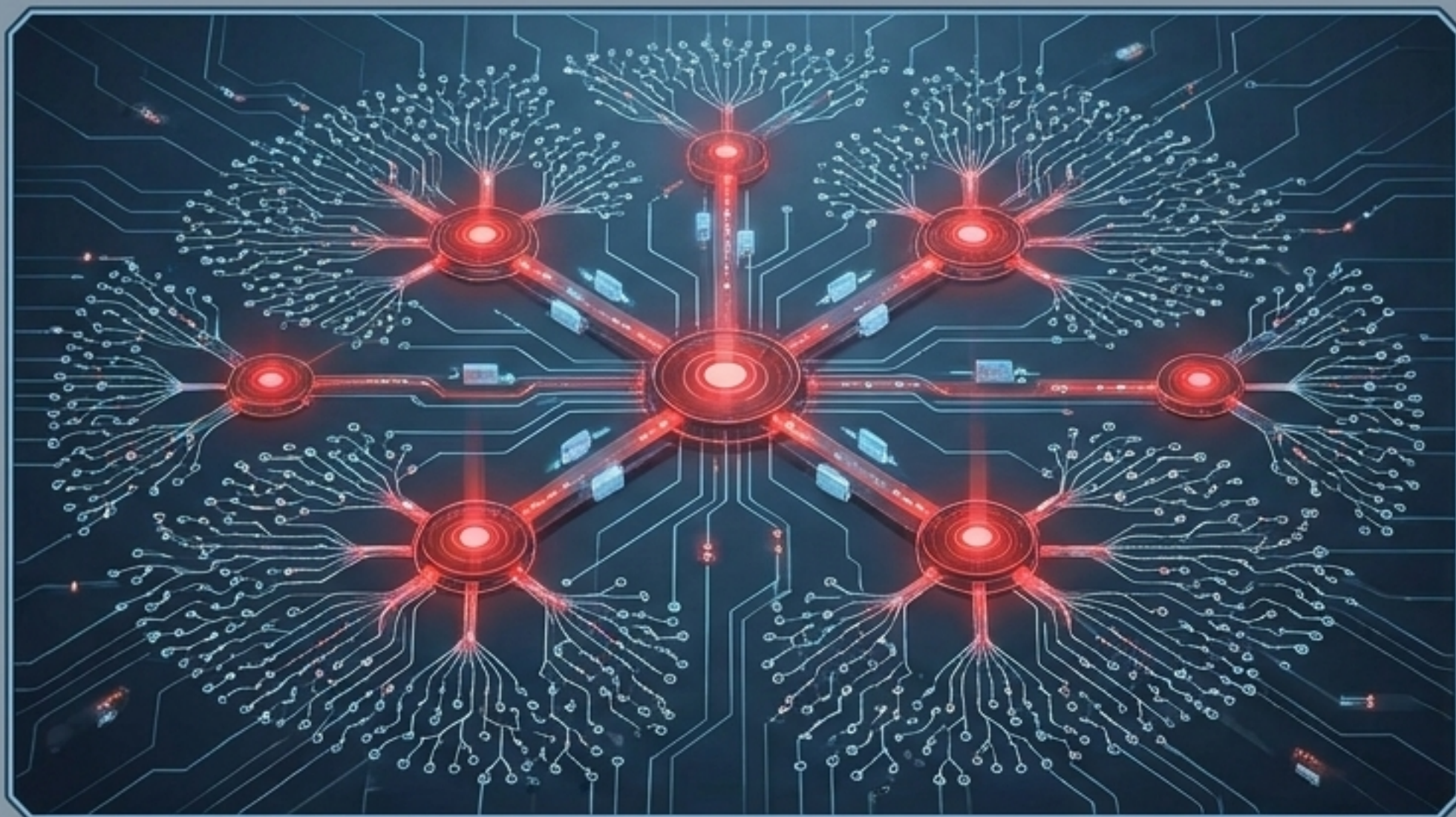
43%

Of all daily flights are direct, rapid responses to critical medical emergencies.

12 Million+
Population Covered

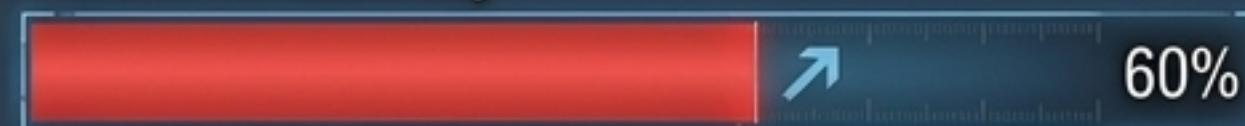
2,300+
Health Facilities Served

600 Deliveries/Day
Capacity per Hub



Performance Diagnostic

Vaccine Availability



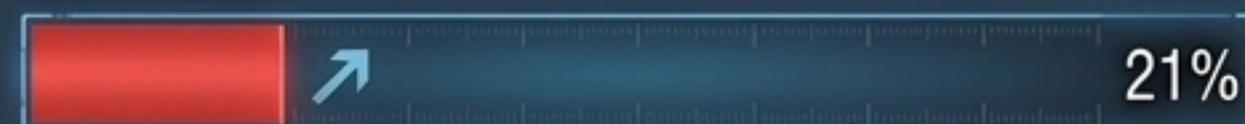
Reduction in facility stock-outs.

Efficiency



Decrease in inventory-driven missed vaccination opportunities.

Resilience



Decrease in total days facilities operated without critical supplies.

Milestone: Delivery of over 1,000,000 COVID-19 vaccine doses directly to clinics utilizing specialized ultra-low temperature cold-chain payload systems.

Permissive Incubation

Operate safely at national scale in Rwanda and Ghana, serving direct public health mandates to build operational maturity.

Massive Data Accumulation

Accrue over 120 million autonomous flight miles and 2 million commercial deliveries, creating an unmatched empirical safety dataset.

Real-World Miles
=
Regulatory Capital

The FAA BEYOND Program

Leverage safety data to gain rigorous FAA Part 135 Air Carrier Certification for Beyond Visual Line of Sight (BVLOS) operations in the US.

Prove Autonomous Safety

Demonstrate flawless operation of Detect-and-Avoid systems, Level 4 autonomy capabilities, and zero third-party safety incidents.

US Commercial Translation: Automating the Suburbs



10-Mile Radius

The US Market Calculus:
90% of the US population lives
within 10 miles of a Walmart.
P2 is explicitly engineered
to capture this radius
silently and autonomously.

Commercial Integration

Retail & E-Commerce (Walmart)

Operations scaling in Dallas-Fort Worth and Pea Ridge, Arkansas. Moving everyday retail from 3,000-lb delivery vans to 55-lb precise aerial drops.

Commercial Integration

Healthcare (Intermountain & Novant)

Delivering prescriptions and essential medical supplies directly to residential doorsteps across Utah and North Carolina.

Commercial Integration

Food Services (Instant Delivery)

Expanding rapid instant delivery with partners like sweetgreen, Chipotle, and Panera, proving multi-vertical platform utility.

Global Telemetry: The Autonomous Impact Dashboard

120,000,000+

Autonomous Commercial
Miles Flown

2,000,000+

Total Commercial
Deliveries

0

Third-Party
Incidents

5,000+

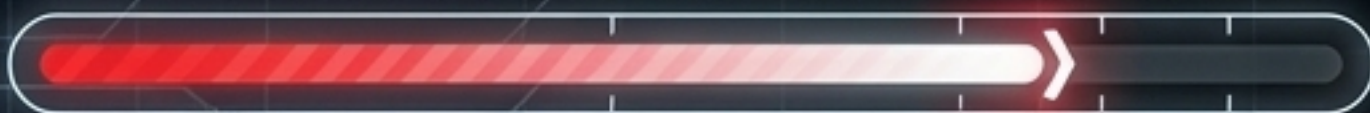
Hospitals and Health Centers
integrated globally.

5,000,000+

Units of vaccines and medical
products delivered.

98%

Reduction in delivery emissions
compared to traditional
automotive logistics.



Production Velocity: By the end of 2025, Zipline manufacturing scaled to produce one new Zip aircraft every hour to meet network expansion demands.

The Uncatchable Moat: Lessons in Instant Logistics

Pillar 1: Hardware-Network Symbiosis



Industry:

Trying to build one compromise drone for all potential use cases.



Zipline:

Deploying specialized hardware for specific network densities: P1 (Catapult/Distance for rural) and P2 (eVTOL/Precision for urban).

Pillar 2: The Data Head Start



Industry:

Testing in limited, highly geofenced US suburban pilot zones.



Zipline:

Over a decade of national-scale operations provides an insurmountable lead in edge-case training and machine learning for flight controllers.

Pillar 3: Regulatory Integration



Industry:

Treating regulators as obstacles and reacting to evolving airspace rules.



Zipline:

Operating as an established, certified air carrier (Part 135) with fully integrated FAA traffic management and onboard Detect-and-Avoid.

Stratosphere Navy

Zipline did not just build a delivery drone; they built a Level-4 autonomous mini-airline that replaces ground infrastructure entirely.